AI readiness of federal statistical publishers: what one measurement shows

Prepared 11 October 2026.

What the framework is for

The framework exists to rate and score, quantitatively and qualitatively, whether the public and the tools the public now relies on can reach, understand and use the data they paid for. The Department of Commerce has set a neighbouring target in its guidance glossary, which defines AI-ready data as material that is not just machine-readable but machine-understandable, enriched with contextual metadata and organized in interpretable standard formats for AI systems to work with. That glossary belongs to guidance from the Department rather than to a Department-wide regulation, and set beside the project's own statement it describes a property of the published material, whereas the project asks whether people and their tools can get to that material.

Publishing a file and reaching it are separate problems, because each step between release and practical value can fail on its own. Wilkinson and colleagues, setting out the FAIR principles, describe an automated agent that must identify an object, judge whether it is useful, judge whether licence, consent and access let it proceed, and only then act, so an agent can clear one of these steps and still stop at the next. Structured metadata on a page therefore does not by itself make the content workable for software, since discovery, access, interpretation and fitness for use are each required, and federal statistical guidance defines quality in the same terms, as fitness for a user's own needs. For example, figures released in aggregated spreadsheets laid out for presentation must be reshaped before analysis even after publication, and a format that is not machine-readable restricts what the posted figures can serve.

Writing for automated readers already has a named discipline, generative engine optimization, which its originators present as the successor to search engine optimization. Those readers face a further barrier, because publishers increasingly restrict AI crawlers, and what the public can reach through AI assistants hinges on those restrictions. The cost of such a restriction falls on the public, since OpenAI says that sites which block its search crawler will be left out of its assistant's search answers, and Anthropic warns that blocking its search crawler may reduce a site's visibility and accuracy in search results.

The framework's structure starts from the 4 criteria USAFacts set out, that data be accessible, understandable, accurate and open, and adds 3 more, for 7 criteria in all. Below the criteria, the framework record holds 50 indicators, of which 48 are in the framework and 2 are candidates. Those 2 candidates enter no count until the project's operator adopts them, and a later section describes what each one tests.

What was measured, on whom, and how

That framework is put to work in one composite measurement cycle, scan_2026-10-07_composite_d, which is the measurement behind this report and covers 16 statistical bodies. Of those bodies, 13 are ranked, because the others have no scored result that rests on a judged finding. Across all of them, the cycle judged 24 sets of results and holds 1,055 findings.

Each of those findings came from a single public client, the scanner, which identified itself by a user agent that names the project and links to its public source. Beyond that name the scanner sent nothing, so no credentials, cookies or other identity accompanied any request. It also kept the same identity for the whole cycle and never varied it to get a better answer.

So far, rules measure only part of the framework, because 23 of its 48 indicators have a current rule. There are 25 current rules in all, but the ones beyond those 23 indicators serve the two checks introduced at the start of this report, one asking whether a permitted machine client is actually served and the other whether it can find its way from the product page, and both stay outside the framework count. The record's own status labels draw the

same line, since it lists 23 of the 48 indicators as measured, 1 whose measuring tool exists but is not yet recorded as measured, and 24 as specified only.

The project's corpus of admitted documents began with a first set of 71 and now holds 282. Of the documents added after that first set, 162 came in before the framework record was first committed and 49 on or after that commit. Those 49 later documents are not all literature, however, because 25 of them are captured copies of the scanned web pages rather than published work.

What was found: hosts that refused a plain client

Turning from the measurement's design to its results, the scanner identified itself and carried no credentials, and on 341 of its 1,055 findings it met a denial of access, in which the host answered with an HTTP 401, 403 or 429 status, disallowed the request in robots.txt, or reset the connection. The remaining outcomes divide into 415 absences, where the thing sought is not served at all, 57 cases where something is served but does not meet the test, 65 other errors, such as failures on the server, responses the scanner could not parse and a redirect loop, and 177 passes.

At the Bureau of Labor Statistics and at the Office of Research, Evaluation, and Statistics of the Social Security Administration, every one of the 45 findings was a denial, so neither body produced a single absence or a single pass. The Bureau of Transportation Statistics came close behind, refusing 41 of its 45 and passing only 2. Bodies with 89 findings each were refused less completely but still often; for example, the Bureau of Economic Analysis turned the scanner away on 40 of them.

At the other end of the range, Statistics of Income at the Internal Revenue Service refused the scanner on only 4 of its 67 findings, and all 4 were robots.txt disallows. That body was not open in any fuller sense, however, because 44 of the 67 are absences and only 12 pass. The National Center for Education Statistics and the National Animal Health Monitoring System each refused 10 of 45. Data.gov is kept apart from these bodies because it is a comparator host and not one of them, and it refused the scanner on 1 of 9 while passing 7.

These denials bear on public access as well as on the scanner, because, as the opening section set out, the public reaches a publisher through AI tools only as far as that publisher's restrictions on crawlers allow.

What was found: what the hosts that answered serve

Where a host answered instead of refusing, the commonest outcome among the 1,055 findings was an absence, a case in which the item a check looked for was not served, and there are 415 of them. Against those, only 57 findings reach something present that falls short of the check, and 177 pass. The rest are the 341 refusals already described and 65 other errors of the kinds listed in the previous section. Absences weigh heavily because, as the opening section set out, use by a machine requires discovery, access, interpretation and fitness for use, each in its turn. Publication alone does not settle the matter either, because data buried in spreadsheets built for presentation is not usable by a machine even when it is published, as that section also noted.

The Census Bureau shows how these results look for one publisher, since its published result tables mark 18 of 39 judged rows as failing. Those tables carry one row for each check and page they report, while the scan holds 89 findings for that publisher across every page it probed, so the tables show less than the full record. Every failing row falls on one of the two pages that present its flagship programs, the surveys and programs page and the American Community Survey page, and the same checks fail on both.

The Census catalog also shows why something the scanner did not find must be kept apart from something that does not exist. The Census Bureau serves a catalog on census.gov holding 1,810 records, but no record in it names the products that were probed. Federal policy expects such a listing, because each agency must keep a public inventory of all its data assets on its own website, and a flagship product is one of those assets. Even so, the 20 Census findings that rest on that catalog, from the checks for variable-level semantics, quality metadata, provenance completeness, no dark data and authority metadata, are recorded as errors rather than failures. The reason is that the Commerce Department's inventory, which the check also names, refused the scanner,

so although the Bureau's organization on catalog.data.gov was read, absence is not established. These findings therefore describe products absent from the catalogs that could be read, and they do not show that the Bureau left those products unlisted.

What the measurement cannot say

The same distinction between an error and a failure bounds the measurement as a whole, because a refused request shows only that the scanner was not served, not that the thing it sought is missing. For example, 20 of the Census Bureau's findings, spread over the tests of variable-level semantics, quality metadata, provenance completeness, no dark data and authority metadata, are errors and not failures. They are errors because no record among the 1,810 in the catalog that census.gov serves names the probed product, and the department's inventory, which the indicator also names, refused the request and could not be read, so although the body's organization on catalog.data.gov was read, absence is not established. More broadly, the scanner, which identified itself and did not log in, met a refusal by the host or a disallow in robots.txt on 24 of the 89 findings for that body. Among its 27 absences, moreover, 4 were judged only after some refused or disallowed probes had been set aside, so those verdicts rest on only part of the evidence.

A body's place in the ranking can rest on far less evidence than its full set of findings would suggest. Under the hierarchical equal-weight score, however, the Bureau ranks 7 of 13 with a score of 0.143, and that position turns on a single test, whether concept definitions are published, versioned and linked from variables, on which 0 of 2 judged rows passed. Were that one verdict reversed, the Bureau would rank 1, so a reader should take a body's place as a statement about a few tests rather than about its readiness as a whole.

The framework is also larger than the part of it this measurement scored. Two further indicators, for access policy coherence and for discoverability from the product page, are candidates, and a candidate enters no framework count until the operator adopts it. What holds back the 25 indicators not yet counted as measured differs by kind: 11 have no standard to measure against, 7 need the cooperation of agencies, 6 need open tooling, 1 is not blocked at all, and none is blocked by funding, so money is not what stands between these indicators and their measurement.

Finally, only part of the framework's evidence can be traced to an exact place in a source. Of 50 indicator evidence cells, 10 carry a pinpoint locator, and of the 154 links that tie an indicator to its evidence, 34 point at a located document. Moreover, 16 indicators have no such link at all, and 84 of the 282 documents admitted to the record are cited through one. A reader can therefore confirm that a source stands behind much of the framework, but can find the exact passage behind an indicator in only 10 cases.

What an agency can do, and what is measured next

With those limits stated, the report turns to what an agency can do, and for the Census Bureau every prescribed fix that the claims name sits in the notional hours effort band and carries no cost. Under equal weights, each fix can raise the bureau's hierarchical score by at most a stated amount, which is the gain if every failing result on the matching indicator passed; because no weighting is asserted, each amount is a bound and not a value. The largest of these amounts comes from the changelog, where carrying a revision class on every entry addresses the check for change legibility and can add at most 0.25. Two pairs of fixes address the conditions under which a machine may reuse the bureau's data. The first pair publishes a machine-readable licence and states that licence as an identifier rather than as a sentence, and both serve the check for license clarity with at most 0.125 each. The second pair declares the terms for AI training and AI input in robots.txt and keeps to the categories and values that the Content Signals Policy defines, and both serve the check for reuse permissions for AI, again with at most 0.125 each.

The remaining fixes concern how the bureau explains its concepts and describes its data to a machine. Three of them serve the test, discussed in the previous section, of whether concept definitions are published and linked from variables: publishing the concept definitions as schema.org defined terms, giving each linked one a code, a set and a definition, and linking them from the variables, each with a ceiling of 0.083. The same ceiling of 0.083

applies to naming the set that each code belongs to, which serves the check for semantic consistency, and to linking the methodology from the product page, which serves the test for methodology legibility. Fixes to the markup on that page carry a ceiling of at most 0.036 each, and embedding JSON-LD that describes the data, typing the existing markup as a Dataset, and making that markup conform to the profile it declares all address the test for structured markup. Publishing a release date and declaring the vintage in the markup, not only in a file header, address the test for timeliness of surface under that same limit.

Tying each measured result to a prescribed fix is not new, because outside-in measurement of an organization's published surface, with a remediation step named for each result, is an existing pattern in tools such as Lighthouse and OpenSSF Scorecard. The difference that the documentation read for this report shows is the effort band, because every fix prescribed here carries a notional one, whereas none of the results in that documentation carries an effort estimate. However, that absence is noted over the sources read and is not established for the literature as a whole, so it marks a difference from the tools examined rather than a finding of novelty.

What the framework measures next depends on what blocks each indicator that still lacks a measured verdict. Of the 25 such indicators, 11 wait on a standard, 7 on the agency's cooperation, 6 on open tooling, 0 on funding, and 1 is not blocked. The next measurements therefore take up the indicators that can be read from outside with public tools and no agency action. For example, the Wayback CDX server can supply the evidence for the indicators of stable identifiers and of regression on vintage transition, AIDRIN can supply it for the readiness metrics baseline, and the SLSA verifier can supply it for release authenticity. The indicator for contract stability compares two releases with oasdiff, so it must wait for a second scan. The check for classification/vintage identity is also within reach, because a standard already admitted names the artifact it would read, and the only missing piece is a collector that this project would write. Each of these indicators is to be measured before the summary is drafted, and none of them requires anything from the agency.

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How this report was written

A language model wrote each section from a fixed set of claims, each of which cites its evidence. A second model, of a different family, then read every sentence against those claims alone; a sentence it did not find supported was cut, never reworded, and code cut any sentence holding a number that was not in a claim it cites. Across the drafts, 14 sentences were cut this way, and the report keeps 85. Before the check was trusted, one false sentence was planted in each of the 6 sections, and the checker caught 6 of 6.